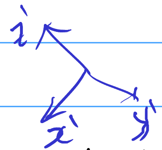
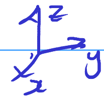
Tensores Cartesianos:



manzana
banana
uva



$$\underline{\Phi}(\underline{x}) = \underline{\Phi}'(\underline{x}')$$

$$u_i' = B_{ij}\, u_j \qquad \underline{u} = \begin{bmatrix} u_1 \\ u_2 \\ u_3 \end{bmatrix}$$

Tensor de rotación del sistema de coordenadas

$$A_{ij}' = B_{ik}\, B_{jl}\, A_{kl} \rightarrow \underline{\underline{A}}' = \underline{\underline{B}}\,\underline{\underline{A}} \cdot \underline{\underline{B}}^{T}$$

Tensor de 2do orden:

$$\underline{u} \cdot \underline{v} = \alpha$$

$$\underline{r} = \underline{\underline{A}} \cdot \underline{n}$$

$$\begin{bmatrix} r_1 \\ r_2 \\ r_3 \end{bmatrix} = \begin{bmatrix} A_{11} & A_{12} & A_{13} \\ A_{21} & A_{22} & A_{23} \\ A_{31} & A_{32} & A_{33} \end{bmatrix} \begin{bmatrix} n_1 \\ n_2 \\ n_3 \end{bmatrix}$$

$$r_1 = A_{11}\, n_1 + A_{12}\, n_2 + A_{13} \cdot n_3 \qquad \underline{n}$$

$$\vdots$$

$$r_3 = A_{31}\, n_1 + A_{32} \cdot n_2 + A_{33}\, n_3$$

→ 2 orden → 2do orden

$$\underline{\underline{\sigma}} = \underline{\underline{\underline{\underline{C}}}} : \underline{\underline{\varepsilon}} \rightarrow \text{8 componentes}$$

↳ 4to orden → $3 \times 3 \times 3 \times 3 = 81$

$$\underline{T} = \underline{\underline{\sigma}} \cdot \underline{n}$$

Tensor de 2 orden